Sector-Aware Multichannel Partition Budgets

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Abstract

We give the exact union and free-energy allocation laws for center, low-dispersion, high-dispersion, cut/orbit, and topology numerators.

Let $B_{V,c}$ be disjoint or merely covering channel numerators and suppose

$$B_{V,c}(s) \le q_c Z_{G,\mathrm{ref}}(s) V(s).$$

Theorem 1 (Typed union law)**.** *The union numerator satisfies*

$$B_V(s) \le \left(\sum_c q_c \right) Z_{G,\mathrm{ref}}(s) V(s).$$

For a target q and positive shares a_c summing to one, the allocation $q_c = a_c q$ is exact. With comparison loss ℓ_c, channel c requires

$$F_c > \ell_c - \log(a_c q).$$

Proof. Subadditivity of integrals gives the first statement. Substitution gives the allocation and free-energy formula. □

For five equal channels, each free-energy requirement is larger than the undivided target by $\log 5$. This prevents five individually plausible bounds from silently overspending one total budget. Ownership labels are retained through the next-state map.

Remark 1. *The allocation theorem is exact. None of the five complete physical channel bounds is yet uniform in volume and retained state.*